

Dental Pulp

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Dental Pulp

Written Task / Draw and label diagrams with captions

1. Composition

- Water: 75%
- Organic Material: 25% (includes collagen fibers, GAGs, proteoglycans, glycoproteins)

2. Collagen Fibers in Pulp

- **Collagen I (60%)** – Main structural support
- **Collagen III (40%)** – Aids in repair and remodeling
- **Fibronectin** – Supports cell adhesion and wound healing
- **Elastin** – Rare; provides limited elastic support

3. Key Functions

- Nutrition: Blood supply from branches of external carotid artery
- Sensation: Via trigeminal nerve (pain, temperature)
- Protection: Secondary/tertiary dentine production
- Defense: Immune cells detect pathogens
- Formation: Stem cells help in repair
- Pressure Regulation: Interstitial fluid maintains balance

4. Pulp Cells

1. Odontoblasts: Form dentine, regulate permeability, immune role
2. Fibroblasts: Produce ECM, support, cell turnover
3. DPSCs: Regenerative potential
4. Defense Cells:
 - Macrophages: Phagocytosis
 - Dendritic Cells: Antigen presentation
 - Lymphocytes: Antibodies

- Mast Cells: Inflammatory response

5. Non-Fibrous Matrix

Glycosaminoglycans (GAGs)

- **Types:** Hyaluronic acid, Chondroitin sulfate, Dermatan sulfate, Heparan sulfate
- **Functions:** Provide hydration, support healing, maintain flexibility, and assist in growth

Proteoglycans

- **Types:** Versican, Syndecan, Decorin, Biglycan
- **Functions:** Aid in tissue repair, regulate collagen formation, and support cellular signaling

Glycoproteins

- **Type:** Fibronectin
- **Functions:** Promote cell adhesion, growth, migration, and wound healing

6. Zones of Dental Pulp

Zone	Contents
Odontoblast Layer	Dentine-forming cells
Cell-Free Zone (Weil)	Few nuclei, axons, capillaries
Cell-Rich Zone	Fibroblasts, immune cells, capillaries, nerves
Pulp Core	Loose CT, blood vessels, nerves

7. MCQ

A researcher studying pulp regeneration after cavity prep needs to target a zone with high cellular activity for stem cell interaction. Which pulp zone is best suited for such regenerative studies?

Options:

- A. Odontoblast Layer
- B. Cell-Free Zone
- C. Cell-Rich Zone
- D. Pulp Core
- E. None of the above

Correct Answer: C. Cell-Rich Zone

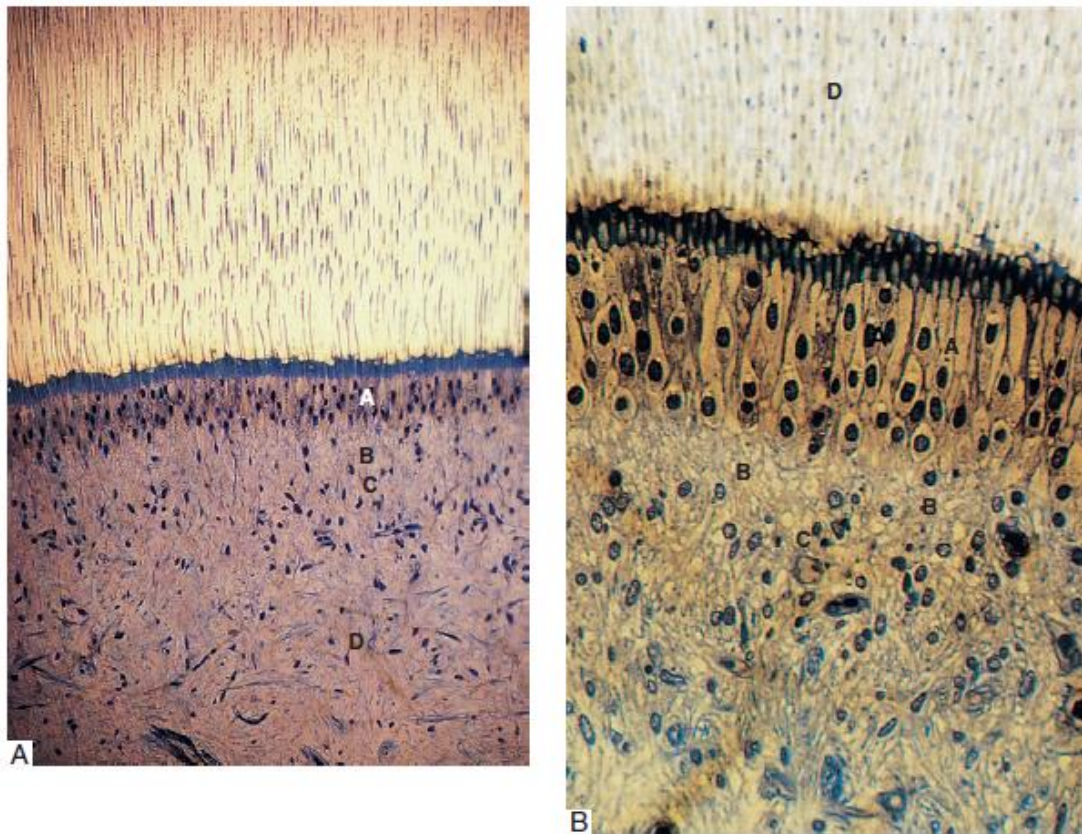


Fig. 10.10 (A) Low-power micrograph showing some cell types and their distribution in the pulp. A = odontoblast layer adjacent to dentine; B = cell-free zone; C = cell-rich zone; D = central pulp showing principally fibroblasts (toluidine blue; $\times 200$). (B) Higher-power view showing pseudostratified odontoblast layer (A). B = cell-free zone; C = cell-rich zone; D = dentine (toluidine blue; $\times 650$).

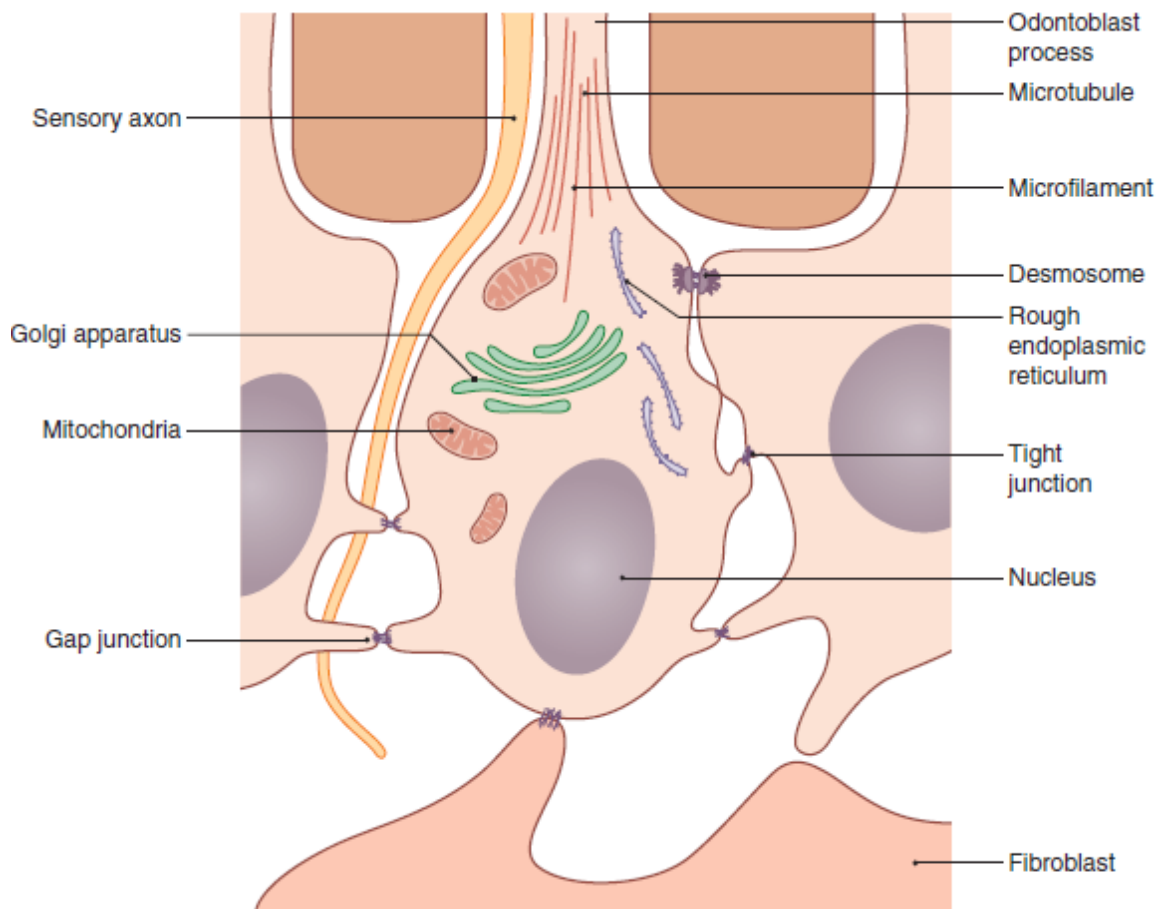


Fig. 10.12 The ultrastructure of the odontoblast in a mature tooth is typical of a polarised protein-synthesising cell of low activity containing all the organelles associated with this process (mitochondria, rough endoplasmic reticulum, Golgi apparatus) in the supranuclear region. One large process enters a dentinal tubule but many smaller ones link the odontoblast to its neighbouring odontoblasts and fibroblasts.